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PASSIVATION METHOD, CELL, AND AUXILIARY PASSIVATION DEVICE FOR IMPLEMENTING PASSIVATION METHOD

Abstract

Disclosed are a passivation method, a cell, and an auxiliary passivation device for implementing the passivation method to solve a problem of a gap between adjacent sheet-like materials when passivating to-be-passivated side surface of the sheet-like materials, causing process gas to wrap-around plating the sheet-like materials along the gap. The passivation method includes: placing a sheet-like material on a carrying device; covering at least one layer of protection film on a surface, away from the carrying device, of the sheet-like material; stacking a next sheet-like material on a side, away from the sheet-like material, of the protection film; and passivating to-be-passivated side surface of stacked sheet-like materials. The wrap-around plate with the process gas on the sheet-like material along the gap between adjacent sheet-like materials during passivation may be avoided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This present disclosure claims priority to Chinese Patent Application No. 202410245192.6, filed on Mar. 4, 2024, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of cell technologies, and in particular, to a passivation method, a cell, and an auxiliary passivation device for implementing a passivation method.

BACKGROUND

[0003] When passivating to-be-passivated side surface of sheet-like materials, a plurality of sheet-like materials are stacked next to each other, with to-be-passivated side surface exposed to the process gas.

[0004] However, a front surface and a back surface of the sheet-like materials include grid lines. The grid lines have a certain height in the thickness direction of the sheet-like materials, resulting in gaps between adjacent sheet-like materials, and causing the process gas to wrap-around plate the sheet-like materials along the gaps between adjacent sheet-like materials. Excess film layers on surfaces not to be passivated of the sheet-like materials are generated, and performance of the sheet-like materials is affected.

SUMMARY

[0005] The present disclosure provides a passivation method. The passivation method includes: placing a sheet-like material on a carrying device, where the carrying device is configured to carry the sheet-like material for bringing it into a passivation device for passivation, the sheet-like material is configured to form a cell, and at least one of a front surface and a back surface of the sheet-like material comprises a grid line; covering at least one layer of protection film on a surface, away from the carrying device, of the sheet-like material; stacking a next sheet-like material on a side, away from the sheet-like material, of the protection film, where a to-be-passivated side surface of the sheet-like material and a to-be-passivated side surface of the next sheet-like material are located on a same side, and the protection film fills a gap between adjacent sheet-like materials; and passivating to-be-passivated side surface of stacked sheet-like materials.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] To make aforementioned as well as other objectives, features, and advantages of the present disclosure more apparent, a more detailed description of embodiments of the present disclosure is given with reference to the drawings. The drawings are intended to provide further understanding of the embodiments of the present disclosure and constitute a part of the specification, used to explain the present disclosure together with the embodiments of the present disclosure, and do not constitute a limitation on the present disclosure. In the drawings, the same reference numerals generally represent the same components or steps.

[0007] FIG. **1** is a structural schematic diagram of stacked sheet-like materials.

[0008] FIG. **2** is a partially enlarged view at A in FIG. **1**.

[0009] FIG. **3** is a flowchart of a passivation method according to an embodiment of the present disclosure.

[0010] FIG. **4** is a schematic diagram of an application scenario applicable to a carrying device according to an embodiment of the present disclosure.

[0011] FIG. **5** is a structural schematic diagram of a protection film and a sheet-like material according to an embodiment of the present disclosure.

[0012] FIG. **6** is a structural schematic diagram of a protection film and a sheet-like material according to another embodiment of the present disclosure.

[0013] FIG. **7** is a flowchart of a passivation method according to another embodiment of the present disclosure.

[0014] FIG. **8** is a flowchart of a passivation method according to still another embodiment of the present disclosure.

[0015] FIG. **9** is a schematic diagram of an application scenario of a carrying device according to an embodiment of the present disclosure.

[0016] FIG. **10**A is a schematic diagram of an application scenario of a carrying device according to another embodiment of the present disclosure.

[0017] FIG. **10**B is an application scenario of a carrying device according to another embodiment of the present disclosure.

[0018] FIG. **11** is a partially enlarged view at B in FIG. **10**A according to an embodiment of the present disclosure.

[0019] FIG. **12** a flowchart of a passivation method according to yet another embodiment of the present disclosure.

[0020] FIG. **13** is a schematic diagram of an application scenario of a carrying device according to still another embodiment of the present disclosure.

[0021] FIG. **14** is a flowchart of a passivation method according to still yet another embodiment of the present disclosure.

[0022] FIG. **15** is a schematic diagram of principle of forming a sheet-like material according to an embodiment of the present disclosure.

[0023] FIG. **16** is a structural schematic diagram of a cell according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0024] The technical schemes in the embodiments of the present disclosure will be described clearly and completely below in combination with the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only a part of the embodiments of the present disclosure, not all of the embodiments. Based on the embodiments in the present disclosure, all other embodiments obtained by those skilled in the art without creative efforts shall fall within the protection scope of the present disclosure.

[0025] FIG. **1** is a structural schematic diagram of stacked sheet-like materials. FIG. **2** is a partially enlarged view at A in FIG. **1**.

[0026] As shown in FIG. **1** and FIG. **2**, when passivating a to-be-passivated side surface **11** of a sheet-like material **10**, a plurality of sheet-like materials **10** are stacked next to each other, and the to-be-passivated side surface **11** is exposed to process gas.

[0027] However, a plurality of grid lines **14** are arranged on a front surface **12** and a back surface **13** of the sheet-like material **10**. The plurality of grid lines **14** extend from the surface of the sheet-like material **10** for forming a gap between adjacent sheet-like materials **10** when stacking the plurality of sheet-like materials **10**, and the sheet-like material **10** would form a wrap-around contact by the process gas introduced into the gap. The performance of the sheet-like material **10** would be influenced if the sheet-like material **10** formed a wrap-around contact.

[0028] To solve the above issues, the present disclosure provides a passivation method, a cell, and an auxiliary passivation device for implementing a passivation method. The following is a detailed description of the specific steps of the passivation method, as well as the specific structure of the cell and the auxiliary passivation device for implementing a passivation method with reference to the drawings and specific embodiments.

[0029] FIG. **3** is a flowchart of a passivation method according to an embodiment of the present disclosure. FIG. **4** is a schematic diagram of an application scenario applicable to a carrying device according to an embodiment of the present disclosure. FIG. **5** is a structural schematic diagram of a protection film and a sheet-like material according to an embodiment of the present disclosure. FIG. **6** is a structural schematic diagram of a protection film and a sheet-like material according to another embodiment of the present disclosure.

[0030] As shown in FIG. **3** to FIG. **6**, the passivation method provided in this embodiment is configured to passivate the to-be-passivated side surface **11** of the plurality of sheet-like materials **10** stacked. The sheet material **10** is used to form cells (not shown in the figure). At one grid line **14** is arranged on the surface of the sheet-like material **10**. The grid line **14** could arranged on a front surface **12** or a back surface **13** of the sheet-like material **10**. It is also possible to arrange one grid line **14** on each of the front surface **12** and the back surface **13**. FIG. **15** is a schematic diagram of principle of forming a sheet-like material according to an embodiment of the present disclosure. As shown in FIG. **15**, the sheet-like material **10** extends approximately along a plane parallel to a first direction X and a second direction Y, and the grid line **14** extends along the first direction X. The to-be-passivated side surface **11** of the sheet-like material **10** is located at an end of the sheet-like material **10** in the first direction X. In other words, the front surface **12** and the back surface **13** extend approximately along the plane parallel to the first direction X and the second direction Y. The to-be-passivated side surface **11** extends approximately along a plane parallel to the second direction Y and a third direction Z. The third direction Z is perpendicular to the first direction X and the second direction Y. The size of the to-be-passivated side surface **11** in the third direction Z is approximately equal to the thickness of the sheet-like material **10**. The sheet-like materials **10** are stacking along the third direction Z. The to-be-passivated side surface **11** is a section plane formed by cutting the original sheet-like material **1**. The sheet-like material **10** is obtained by cutting the original sheet-like material **1** along the dashed line in FIG. **15**. The sheet-like material **10** includes a front surface **12** (i.e., upper surface), a back surface **13** (i.e., lower surface), a to-be-passivated side surface **11**, and a non-passivated side surface **15**.

[0031] The passivation method includes the following steps.

[0032] Step S**10**: Placing a sheet-like material **10** on a carrying device **20**.

[0033] The carrying device **20** is configured to carry the sheet-like material **10** for bringing it into a passivation device **30** for passivation.

[0034] Specifically, the sheet-like material **10** may include a silicon wafer, a wafer, a cell, and so on. The sheet-like material **10** may be a raw material or semi-manufactured product configured to form a cell. The front surface **12** of the sheet-like material **10** may include a side surface, coated with an anti-reflective layer, of the silicon wafer. The back surface **13** of the sheet-like material **10** may include a surface opposite to the front surface **12**. In some embodiments, the front surface **12** of the sheet-like material **10** is not provided with a grid line, for example, the sheet-like material **10** configured to form a back contact battery.

[0035] The carrying device **20** may be any device capable of carrying the sheet-like material **10**. As shown in FIG. **4**, the carrying device **20** may further include a device for carrying the sheet-like material **10** when passivating the to-be-passivated side surface **11** in the passivation device **30**. The passivation device **30** may include a passivation reaction furnace, a vapor deposition device, and so on.

[0036] Step S**20**: Covering at least one layer of protection film **40** on a surface of the sheet-like material **10**, away from the carrying device **20**.
[0037] Specifically, the protection film **40** may be a membranous object. The protection film **40** may be covered on the surface of the sheet-like material **10** by placing, coating, laying, and so on.
[0038] Step S**30**: Stacking a next sheet-like material **10** on a side, away from the sheet-like material **10**, of the protection film **40** corresponding to the sheet-like material **10** covered in step S**20**. The to-be-passivated side surface **11** of the sheet-like material **10** and the to-be-passivated side surface **11** of the next sheet-like material **10** are located on a same side.
[0039] Specifically, one or more layers of protection film **40** may be disposed between adjacent sheet-like materials **10**. The protection film **40** may fill or cover a gap between adjacent sheet-like materials **10**. For example, as shown in FIG. **5** and FIGS. **6**, the protection film **40** includes a protection film body **41**, and at least one groove **42** on the side contacted with the sheet-like material **10**. The groove **42** may be configured to accommodate the grid line **14** on the side, contacted with the protection film **40**, of the sheet-like material **10**. The protection film body **41** may fill or cover the gap between adjacent sheet-like materials **10**.
[0040] Exemplarily, the surface, which includes the grid lines **14** and is configured to contact the sheet-like material **10**, of the protection film **40** includes a plurality of grooves **42**. The number of the grooves **42** is equal to the number of the grid lines **14** in contact with the sheet-like material **10**. The shape of the groove **42** is adapted to the shape of the grid line **14**. For example, the shape of the grid line **14** is a long strip, and the shape of the groove **42** is a long strip. The width of the groove is equal to the width of the grid line **14**, the depth of the groove is equal to the height of the grid line **14**, and the length of the groove **42** is equal to the length of the grid line **14**.
[0041] Exemplarily, the surface of the protection film **40** may have no grooves **42**. Specifically, the material of the protection film **40** may be produced by an elastic material. After a surface of the sheet-like material **10** having the grid line **14** is in contact with the protection film **40**, the grid lines **14** may be compressed against the protection film **40** so that the grid lines **14** are embedded in the protection film **40**. In other words, before the protection film **40** contacts with the sheet-like material **10**, there may be no grooves on the surface of the protection film **40**; however, after the protection film **40** contacts with the sheet-like material **10**, the grid lines **14** are embedded in the protection film **40**, forming the grooves **42** on the protection film **40**.
[0042] Exemplarily, a plurality of protection films **40** are disposed between adjacent sheet-like materials **10**, and the shape of the protection films **40** may be the same or different. For example, the surface of the sheet-like material **10**, away from the carrying device **20**, may be either the front surface **12** or the back surface **13** of the sheet-like material **10**. The area of the protection film **40** may be greater than or equal to the area of the front surface **12** or the back surface **13** of the sheet-like material **10**.
[0043] Step S**40**: Passivating to-be-passivated side surface **11** of the sheet-like materials **10** stacked.
[0044] Exemplarily, the to-be-passivated side surface **11** of each of the sheet-like materials **10** may include a cross section or a cut section generated on the side of the sheet-like material **10** when forming the sheet-like material **10**.
[0045] Exemplarily, for the sheet-like materials **10** stacked named as a stacked sheet-like materials **10**, at least one layer of protection film **40** may be set between adjacent sheet-like materials **10** with or without the grid lines **14** on the opposite surfaces of the adjacent sheet-like materials **10**. In some embodiments, the stacked sheet-like materials **10** may include stacking the front surfaces **12** of the stacked sheet-like materials **10** in the same direction. In some embodiments, the stacked sheet-like materials **10** may include that the stacked sheet-like materials **10** are horizontally stacked.
[0046] According to the passivation method provided in this embodiment, by providing the protection film **40** between the adjacent sheet-like materials **10**, which fills or covers the gap between adjacent sheet-like materials **10**, the wrap-around plate with the process gas on the sheet-

like material **10** along the gap between adjacent sheet-like materials **10** during passivation may be avoided, thereby improving the performance of the passivated sheet-like materials **10**.

[0047] FIG. **7** is a flowchart of a passivation method according to another embodiment of the present disclosure. The embodiment shown in FIG. **7** is an extension of the embodiment shown in FIG. **3**, and differences between the embodiment shown in FIG. **7** and the embodiment shown in FIG. **3** are introduced bellow, and the similarities are not repeated.

[0048] The carrying device **20** includes a first flat plate. The first flat plate may include a plate-shaped component. The first flat plate may be configured to carry the stacked sheet-like materials **10** when passivating the to-be-passivated side surface **11** in the passivation device **30**. Exemplarily, when passivating the to-be-passivated side surface **11** in the passivation device **30**, the first flat plate is disposed parallel to the sheet-like materials **10** and is located at the bottom of the stacked sheet-like materials **10**. Exemplarily, the side, near the sheet-like material **10**, of the first flat plate may include a plane.

[0049] As shown in FIG. **7**, before the step S**10**: Placing a sheet-like material **10** on a carrying device **20**, the passivation method further includes step S**50**: Covering the protection film **40** on a side of the first flat plate.

[0050] Step S**10**: Placing a sheet-like material **10** on a carrying device **20**, includes step S**11**: Stacking the sheet-like material **10** on a side of the protection film **40**, away from the first flat plate.

[0051] Exemplarily, the surface, near the first flat plate, of the sheet-like material **10** may be provided with at least one of the grid lines **14**.

[0052] Step S**20**: Covering at least one layer of protection film **40** on a surface of the sheet-like material **10**, away from the carrying device **20**, includes step S**21**: Covering at least one layer of the protection film **40** on a surface of the sheet-like material **10** away from the first flat plate. Step S**30**: Stacking a next sheet-like material **10** on a side, away from the sheet-like material **10**, of the protection film **40** corresponding to the sheet-like material **10** covered in step S**20**, is further configured as: Stacking a next sheet-like material **10** on a side, away from the sheet-like material **10**, of the protection film **40** corresponding to the sheet-like material **10** covered in step S**21**.

[0053] According to the passivation method provided in this embodiment, the carrying device **20** includes the first flat plate. When passivating the to-be-passivated side surface **11** in the passivation device **30**, the first flat plate may carry the sheet-like material **10**. By disposing the protection film **40** between the first flat plate and the sheet-like material **10** adjacent to the first flat plate, the protection film **40** may fill or cover the gap between the first flat plate and the sheet-like material **10** adjacent to the first flat plate, thereby avoiding the process gas from wrap-around plating the sheet-like material **10** along the gap between the first flat plate and the sheet-like material **10** adjacent to the first flat plate during passivation of the to-be-passivated side surface **11** of the sheet-like material **10**, thereby further improving the performance of the passivated sheet-like materials **10**.

[0054] In some embodiments, before step S**40**: Passivating to-be-passivated side surface **11** of stacked sheet-like materials **10**, the passivation method further includes: Covering at least one layer of the protection film **40** on a side, away from the first flat plate, of the sheet-like material **10** farthest from the first flat plate. This may prevent the surface of the sheet-like material **10**, which is farthest from the first flat plate, from being plated with process gas on the side away from the first flat plate.

[0055] FIG. **8** is a flowchart of a passivation method according to still another embodiment of the present disclosure. The embodiment shown in FIG. **8** is an extension of the embodiment shown in FIG. **7**, and differences between the embodiment shown in FIG. **8** and the embodiment shown in FIG. **7** are introduced bellow, and the similarities are not repeated. FIG. **9** is a schematic diagram of an application scenario of a carrying device according to an embodiment of the present disclosure.

[0056] As shown in FIG. **9**, the carrying device **20** further includes a second flat plate **22**. The second flat plate **22** may be configured to be sent into the passivation device **30** together with the sheet-like material **10**. The second flat plate **22** may include a plate-shaped component. The side, near the sheet-like material **10**, of the second flat plate **22** may include a plane. In FIG. **9**, in order to display the second flat plate **22**, the sheet-like material **10**, and the protection film **40**, the second flat plate **22**, partial sheet-like material **10**, and partial protection film **40** are displayed in the form of an exploded view. In practical applications, the second flat plate **22**, the sheet-like material **10**, and the protection film **40** are compressed.

[0057] As shown in FIG. **8**, before step S**40**: Passivating to-be-passivated side surface **11** of stacked sheet-like materials **10**, the passivation method further includes the following steps.

[0058] Step S**60**: Covering at least one layer of the protection film **40** on a side, away from the first flat plate **21**, of the last sheet-like material **10**.

[0059] Exemplarily, the last sheet-like material **10** may be the sheet-like material **10** farthest from the first flat plate **21**.

[0060] Step S**70**: Stacking the second flat plate **22** on a side, away from the first flat plate **21**, of the last protection film **40**.

[0061] Exemplarily, the last protection film **40** may be the protection film **40** on the side, away from the first flat plate **21**, of the last sheet-like material **10**.

[0062] The second flat plate **22** is disposed opposite to the first flat plate **21** to form an accommodation space. The accommodation space is configured to accommodate the stacked sheet-like materials **10**. Exemplarily, the second flat plate **22** is disposed parallel to the sheet-like material **10** and the first flat plate **21**. Exemplarily, the second flat plate **22** may be configured to compress the staked sheet-like materials **10** and the protection film **40**, or to cooperate with the first flat plate **21** to compress the staked sheet-like materials **10** and the protection film **40**, so that the first flat plate **21**, the second flat plate **22**, and the sheet-like material **10** and the protection film **40** between the first flat plate **21** and the second flat plate **22** are in close contact, avoiding the process gas from wrap-around plating the sheet-like material **10** along the gap between the first flat plate **21**, the second flat plate **22**, and the sheet-like material **10** between the first flat plate **21** and the second flat plate **22**, thereby further improving the performance of the passivated sheet-like materials **10**.

[0063] According to the passivation method provided in this embodiment, the second flat plate **22** and the first flat plate **21** form the accommodation space to accommodate the stacked sheet-like materials **10**; and the protection film is disposed between the last sheet-like material **10** and the second flat plate **22**, the protection film **40** may fill or cover the gap between the last sheet-like material **10** and the second flat plate **22**, avoiding the process gas from wrap-around plating the sheet-like material **10** through the gap between the sheet-like material **10** and the second flat plate **22**, thereby further improving the performance of the passivated sheet-like material **10**.

[0064] FIG. **10**A is a schematic diagram of an application scenario of a carrying device according to another embodiment of the present disclosure. FIG. **10**B is an application scenario of a carrying device according to another embodiment of the present disclosure. FIG. **11** is a partially enlarged view at B in FIG. **10**A.

[0065] As shown in FIG. **10**A and FIG. **10**B, the carrying device **20** includes: a first flat plate **21**, a second flat plate **22**, and a film box **23**. The film box **23** includes a box body **230** and a box cover **231**. The box cover **231** is detachably connected to the box body **230**. The box body **230** includes an accommodation chamber **2301**, and the accommodation chamber **2301** is configured to accommodate the first plate **21**, the second plate **22**, and a sheet-like material **10** and a protection film **40** accommodated between the first flat plate **21** and the second flat plate **22**.

[0066] As shown in FIG. **10**B, a gap **2302** is provided between the box body **230** and the box cover **231** when the box body **230** is fastened with the box cover **231**. The gap **2302** is configured as an inlet for process gas. When the passivation device **30** passivates the to-be-passivated side surface **11**, the process gas enters the film box through the gap **2302**. When the sheet-like material **10** is accommodated in the accommodation chamber, the to-be-passivated side surface **11** of the sheet-like material **10** is placed towards the side, where the gap **2302** is disposed, of the film box **23**.

[0067] In this embodiment, the sheet-like material **10** further includes at least one non-passivated side surface **15**. The non-passivated side surface **15** includes the side of the sheet-like material **10** that does not need to be passivated. The box body **230** is composed of a plurality of box boards **2300**. The box board **2300** of the box body **230** is located on a side facing the non-passivated side surface **15** of the sheet-like material **10**, so that the box board **2300** may cover the non-passivated side surface **15** of the sheet-like material **10**, which may avoid the process gas from entering the box body **230** through the gap **2302** and wrap-around plating the non-passivated side surface **15** of the sheet-like material **10**.
[0068] As shown in FIG. **10**A, FIG. **10**B and FIG. **11**, the box body **230** includes a protrusion **2301** on a side, which is configured to fasten with the box cover **231**, of the box body **230**. The protrusion **2301** may create the gap **2302** when the box body **230** and the box cover **231** is fastened; and a size of the protrusion **2301** in a direction perpendicular to the box cover **231** is equal to a size of the gap **2302** in the direction perpendicular to the box cover **231**. The amount of the process gas entering the film box **23** may be adjusted by adjusting the width of the gap **2302**. In some embodiments, there is a gap between the non-passivated side surface **15** of the sheet-like material **10** and the box board **2300**; by reducing the width of the gap **2302** between the box body **230** and the box cover **231** when they are fastened, the wrap-around plate with the process gas on the non-passivated side surface **15** through the gap between the non-passivated side surface **15** of the sheet-like material **10** and the box board **2300**.
[0069] In some embodiments, before step S**40**: Passivating to-be-passivated side surface **11** of stacked sheet-like materials **10**, the passivation method further includes: sending the carrying device **20** with the stacked sheet-like materials **10** into the passivation device **30**.
[0070] FIG. **12** a flowchart of a passivation method according to yet another embodiment of the present disclosure. The embodiment shown in FIG. **12** is an extension of the embodiment shown in FIG. **3**, and differences between the embodiment shown in FIG. **12** and the embodiment shown in FIG. **3** are introduced bellow, and the similarities are not repeated.
[0071] In this embodiment, the sheet-like material **10** is further provided with at least one non-passivated side surface **15**.
[0072] Step S**20**: Covering at least one layer of protection film **40** on a surface, away from the carrying device **20**, of the sheet-like material **10**, includes step S**22**: Covering the protection film **40** on the surface, away from the carrying device **20**, of the sheet-like material **10** and at least one non-passivated side surface **15**. Step S**30**: Stacking a next sheet-like material **10** on a side, away from the sheet-like material **10**, of the protection film **40** corresponding to the sheet-like material **10** covered in step S**20**, is further configured as: Stacking a next sheet-like material **10** on a side, away from the sheet-like material **10**, of the protection film **40** corresponding to the sheet-like material **10** covered in step S**22**.
[0073] Exemplarily, the protection film **40** may cover the non-passivated side surface **15** of one sheet-like material **10** or adjacent sheet-like materials **10**. For example, as shown in FIG. **5**, the protection film **40** includes an extension portion **410** in a thickness direction of the protection film **40** at the edge of the protective film body **41**. The size, in a thickness direction of the protection film **40**, of the extension portion **410** is greater than or equal to a thickness of at least one sheet-like material **10**. The extension portion **410** of the protection film **40** may cover the non-passivated side surface **15** of the sheet-like material **10**.
[0074] According to the passivation method provided in this embodiment, the protection film **40** may cover the surface, away from the carrying device **20**, of the sheet-like material **10**, as well as at least one non-passivated side surface **15**. That is, the protection film may further cover the non-passivated side surface **15** of the sheet-like material **10**, avoiding the process gas from wrap-around plating the non-passivated side surface **15** of the sheet-like material **10**, and is conducive to simplifying the structure of the carrying device **20** and reducing the load of a device for transporting the carrying device **20**. Exemplarily, the box board **2300**, facing the non-passivated

side surface **15** of the sheet-like material **10**, of the box body **230** may not be disposed. In some application scenarios, when the passivation device **30** passivates the to-be-passivated side surface **11**, it is necessary to preheat the sheet-like material **10**, and the preheating time for the sheet-like material **10** may be shorten by simplifying the structure of the carrying device **20**, thereby improving production efficiency.

[0075] FIG. **13** is a schematic diagram of an application scenario of a carrying device according to still another embodiment of the present disclosure.

[0076] FIG. **14** is a flowchart of a passivation method according to still yet another embodiment of the present disclosure. The embodiment shown in FIG. **14** is an extension of the embodiment shown in FIG. **8**, and differences between the embodiment shown in FIG. **14** and the embodiment shown in FIG. **8** are introduced bellow, and the similarities are not repeated.

[0077] As shown in FIG. **13**, the carrying device **20** further includes a side plate **24**.

[0078] Before step S**40**: Passivating to-be-passivated side surface **11** of stacked sheet-like materials **10**, the passivation method further includes step S**80**: Connecting the side plate **24** to the first flat plate **21** and the second flat plate **22** separately.

[0079] At least one of the first flat plate **21** and the second flat plate **22** is supported by the side plate **24**. When the carrying device **20** with several sheet-like materials **10** is moved, relative movement between the first flat plate **21** and the second flat plate **22** may be prevented. When relative movement occurs between the protection film **40** and the sheet-like material **10**, the damage caused by the protective film **40** to the gap filling or blocking between the first flat plate **21**, the multi sheet-like material **10**, and the second flat plate **22** is prevented, and the structural stability of the carrying device **20** is improved.

[0080] Exemplarily, the side plate **24** is detachably connected to at least one of the first flat plate **21** and the second flat plate **22**. The shape of the side plate **24** may be set according to requirements. In some embodiments, the side plate **24** may be configured to cover the non-passivated side surface **15** when passivating the to-be-passivated side surface **11** of the sheet-like material **10**. In other embodiments, the first plate **21**, the second plate **22**, the side plate **24**, and the stacked sheet-like material **10** are all fed into the film box **23** and carried into the passivation device **30** by the film box **23**.

[0081] In some embodiments, the protection film **40** may cover the surface, away from the carrying device **20**, of the sheet-like material **10**, as well as all non-passivated side surface **15**. Due to the non-passivated side surface of the sheet-like material **10** being protected by the protection film **40**, as shown in FIG. **13**, the carrying device **20** may only include the first flat plate **21**, the second flat plate **22**, and the side plate **24** connected to the first flat plate **21** and the second flat plate **22** separately, and the film box **23** is not necessary. This may further simplify the structure of the carrying device **20**, reduce the load on the device for transporting the carrying device **20**, shorten the preheating time of the passivation device **30** for the sheet-like material **10**, and improve production efficiency. In this embodiment, the side plate **24** does not need to cover the non-passivated side surface **15**. The shape of the side plate **24** may be rod shaped, column shaped, plate shaped, and so on, as long as it may support at least one of the first flat plate **21** and the second flat plate **22**.

[0082] In some embodiments, the to-be-passivated side surface **11** of the sheet-like material **10** and the to-be-passivated side surface **11** of the next sheet-like material **10** are located on a same plane. This setting may avoid the differences in passivation conditions caused by the irregular of the sheet-like material **10**, which is beneficial for improving the uniformity of passivation effect, and may also avoid the situation where the protection film **40** does not fully cover the front surface or the back surface of the sheet-like material **10**, leading to wrap-around plate.

[0083] Exemplarily, the to-be-passivated side surface **11** of stacked sheet-like materials **10** may all be located on the same plane.

[0084] In some embodiments, the protection film **40** includes a film layer **43** with elasticity in a thickness direction to have the grid line **14** of the sheet-like material **10** embedded in the protection film **40**. Since the protection film **40** includes the film layer **43** with elasticity in the thickness direction, as shown in FIG. **5**, the adjacent sheet-like materials **10** may compress and deform the protection film **40**, so that the grid line **14** of the sheet-like material **10** may be embedded in the protection film **40**, allowing the protection film **40** to fill or cover the gap between adjacent sheet-like materials **10**. Exemplarily, when passivating the to-be-passivated side surface **11** of the sheet-like material **10**, the protection film **40** covering the non-passivated surface of the sheet-like material **10** may be compressed by the sheet-like material **10** and the sheet-like material **10** stacked above it.

[0085] In some embodiments, the protection film **40** may further include a flexible film layer. The flexible film layer is easy to be bent and shaped, and it is easy to fit surfaces at different angles. When covering the surface away from the carrying device **20** and at least one side surface not to be passive **15** of the sheet-like material **10**, it may better fit the side surface not to be passive **15**. Exemplarily, the flexible film covering the non-passivated side surface **15** may be naturally sagging or the edge may be clamped. In some embodiments, the protection film **40** may have elasticity in the thickness direction and be flexible.

[0086] In some embodiments, the material of the protection film includes one or more combinations of polyimide, mica polybenzimidazole, and glass fiber.

[0087] The above provides a detailed description of the embodiments of the passivation method, and the following would describe in detail an embodiment of a cell. It should be understood that the description of the embodiments of the passivation method corresponds to the description of the embodiment of the cell. Therefore, the parts that are not described in detail may refer to the embodiments of the passivation method above.

[0088] An embodiment of the present disclosure provides a cell. FIG. **16** is a structural schematic diagram of a cell according to an embodiment of the present disclosure. As shown in FIG. **16**, the cell **60** is obtained based on a sheet-like material **10** prepared by a passivation method according to any of the above embodiments. Exemplarily, the cell **60** includes a photovoltaic cell.

[0089] During the preparation process of the cell provided in this embodiment, a protection film **40** is disposed between adjacent sheet-like materials **10**, which fills or covers the gap between the adjacent sheet materials **10**, thereby avoiding the process gas from wrap-around plating the sheet-like material **10** along the gap between the adjacent sheet materials **10**, thereby improving the performance of the passivated sheet-like materials **10**, and further improving the performance of the prepared cell.

[0090] The above provides a detailed description of the embodiments of the passivation method, and the following would describe in detail an embodiment of an auxiliary passivation device for implementing a passivation method. It should be understood that the description of the embodiments of the passivation method corresponds to the description of the embodiment of the auxiliary passivation device for implementing a passivation method. Therefore, the parts that are not described in detail may refer to the embodiments of the passivation method above.

[0091] An embodiment of the present disclosure provides an auxiliary passivation device for implementing a passivation method. The auxiliary passivation device for implementing a passivation method is applied to implement a passivation method described in any of the above embodiments. The passivation method is configured to passivate the to-be-passivated side surface **11** of stacked sheet-like materials **10**. The sheet-like material **10** is configured to prepare a cell. At least one of the front surface and back surface of the sheet-like material **10** includes a grid line **14**.

[0092] The auxiliary passivation device for implementing a passivation method includes a carrying device **20** and at least one layer of protection film **40**. The carrying device **20** is configured to carry the sheet-like material **10** for bringing it into a passivation device **30** for passivation. And at least one layer of protection film **40** is configured to fill a gap between adjacent sheet-like materials **10**.

[0093] In the application of the auxiliary passivation device for implementing a passivation method provided in this embodiment, the protection film **40** is disposed between the adjacent sheet-like materials **10**, which fills or covers the gap between the adjacent sheet materials **10**, thereby avoiding the process gas from wrap-around plating the sheet-like material **10** along the gap between the adjacent sheet materials **10**, thereby improving the performance of the passivated sheet-like materials **10**, and further improving the performance of the prepared cell.

[0094] Exemplarily, the carrying device **20** may further include a device for carrying the sheet-like material **10** when passivating the to-be-passivated side surface **11** in the passivation device **30**. Exemplarily, the carrying device **20** may carry one or more sets of stacked protection films **40** and sheet-like materials **10**.

[0095] In some embodiments, as shown in FIG. **13**, the sheet-like material **10** is further provided with at least one non-passivated side surface **15**. And the number of the protection film **40** may be multiple.

[0096] The carrying device **20** includes a first flat plate **21**, a second flat plate **22**, and a side plate **24**. The first flat plate **21** is configured to carry the sheet-like material **10**. The second flat plate **22** is disposed opposite to the first flat plate **21** to form an accommodation space. The accommodation space is configured to accommodate the stacked sheet-like materials **10**. The side plate **24** is detachably connected to the first flat **21** plate and the second flat plate **22**.

[0097] The plurality of protection films **40** are disposed respectively between the first flat plate **21** and the sheet-like material **10**, between the adjacent sheet-like materials **10**, and between the sheet-like material **10** and the second flat plate, and covers at least one non-passivated side surface.

[0098] Exemplarily, the side plate **24** is detachably connected to at least one of the first flat plate **21** and the second flat plate **22**. The disassembly and assembly of the side plate **24** with at least one of the first flat plate **21** and the second flat plate **22** may be achieved through manual or automated equipment. Exemplarily, as shown in FIG. **13**, the side plate **24** includes threaded holes **240** on the surface connected to the first flat plate **21** and the second flat plate **22**. The first flat plate **21** and the second flat plate **22** include through holes **50**. The side plate **24**, the first flat plate **21**, and the second flat plate **22** may be fixed by screws (not shown in the figure) through the through holes **50** and the threaded holes, and further, the first flat plate **21** and the second flat plate **22** may be tightened to press the protection film **40** and the sheet-like material **10**. Exemplarily, a spring is disposed between the side plate **24**, the first flat plate **21**, and the second flat plate **22**, which are tightened by the spring, and the first flat plate **21** and the second flat plate **22** are pressed against the protection film **40** and the sheet-like material **10**. Exemplarily, the side plate **24** may be configured to provide a guide groove in the thickness direction of the sheet-like material **10**, so that the first flat plate **21** or the second flat plate **22** slides along the guide groove; for example, the second flat plate **22** is located above the first flat plate **21**, and the protection film **40** and the sheet-like material **10** are compressed through the self-weight of the second flat plate **22**.

[0099] The carrying device **20** provided in this embodiment includes high structural stability, making it easy to send the sheet-like material **10** into the passivation device **30** for passivation. It may also prevent relative movement between the first flat plate **21** and the second flat plate **22** when the carrying device **20** carrying sheet-like materials **10** is moved, causing relative movement between the protection film **40** and the sheet-like material **10**, and damaging the filling or covering of the gaps between the protection film **40** and the first flat plate **21**, the sheet-like materials **10**, and the second flat plate **22**.

[0100] Additionally, while the various steps of the method in the present disclosure are described in a particular order in the drawings, this does not require or imply that these steps must be performed in that specific order, or that all shown steps must be executed to achieve the desired results. Alternatively, some steps may be omitted, multiple steps may be combined into one step, and/or one step may be divided into multiple steps, and so on.

[0101] The basic principles of the present disclosure have been described above with specific embodiments. However, it should be noted that the advantages, benefits, and effects mentioned in the present disclosure are mere examples and not limitations. They should not be taken as essential features that must be possessed by each embodiment of the present disclosure. Additionally, the specific details disclosed above serve merely as illustrations and aids for understanding, and do not constitute limitations. These details do not restrict the necessity of implementing the present disclosure with the aforementioned specific details.

[0102] The block diagrams of devices, apparatuses, equipment, and systems involved in the present disclosure are provided as illustrative examples only and do not require or imply that the connections, arrangements, or configurations must be made in the manner shown in the block diagrams. As will be recognized by those skilled in the art, these devices, apparatuses, equipment, and systems may be connected, arranged, and configured in any manner. Terms such as “include”, “contain”, “have”, and so on are open-ended terms that mean “including but not limited to” and may be used interchangeably. The term “such as” used here refers to the phrase “such as but not limited to” and may be used interchangeably.

[0103] It should also be noted that in the devices, equipment, and methods of the present disclosure, various components or steps may be decomposed and/or recombined. Such decomposition and/or recombination should be considered as equivalent solutions of the present disclosure.

[0104] The above description of the disclosed aspects is provided to enable those skilled in the art to make or use the present disclosure. Various modifications to these aspects will be readily apparent to those skilled in the art, and the general principles defined here may be applied to other aspects without departing from the scope of the present disclosure. Therefore, the present disclosure is not intended to be limited to the aspects shown here, but should be given the widest scope consistent with the principles and novel features disclosed here.

[0105] The above description has been given for the purpose of illustration and description. Furthermore, this description is not intended to limit the embodiments of the present disclosure to the forms disclosed here. Although multiple example aspects and embodiments have been discussed above, those skilled in the art will recognize certain variations, modifications, alterations, additions, and sub-combinations thereof.

Claims

1. A passivation method, comprising: placing a sheet-like material on a carrying device, wherein the carrying device is configured to carry the sheet-like material for bringing it into a passivation device for passivation, the sheet-like material is configured to form a cell, and at least one of a front surface and a back surface of the sheet-like material comprises a grid line; covering at least one layer of protection film on a surface, away from the carrying device, of the sheet-like material; stacking a next sheet-like material on a side, away from the sheet-like material, of the protection film, wherein a to-be-passivated side surface of the sheet-like material and a to-be-passivated side surface of the next sheet-like material are located on a same side, and the protection film fills a gap between adjacent sheet-like materials; and passivating to-be-passivated side surface of stacked sheet-like materials.

2. The passivation method according to claim 1, wherein the carrying device comprises a first flat plate; before the placing a sheet-like material on a carrying device, the passivation method further comprises: covering the protection film on a side of the first flat plate; wherein the placing a sheet-like material on a carrying device comprises: stacking the sheet-like material on a side, away from the first flat plate, of the protection film; and wherein the covering at least one layer of protection film on a surface, away from the carrying device, of the sheet-like material comprises: covering at least one layer of the protection film on a surface, away from the first flat plate, of the sheet-like material.

3. The passivation method according to claim 2, wherein the carrying device further comprises a

second flat plate; and before the passivating to-be-passivated side surface of stacked sheet-like materials, the passivation method further comprises: covering at least one layer of the protection film on a side, away from the first flat plate, of the last sheet-like material; and stacking the second flat plate on a side, away from the first flat plate, of the protection film, wherein the last protection film is the protection film on the side, away from the first flat plate, of the last sheet-like material.
4. The passivation method according to claim 1, wherein the sheet-like material further comprises at least one non-passivated side surface; and the covering at least one layer of protection film on a surface, away from the carrying device, of the sheet-like material comprises: covering the protection film on the surface, away from the carrying device, of the sheet-like material and the at least one non-passivated side surface.
5. The passivation method according to claim 3, wherein the carrying device further comprises a side plate; and before the passivating to-be-passivated side surface of stacked sheet-like materials, the passivation method further comprises: connecting the side plate to the first flat plate and the second flat plate separately.
6. The passivation method according to claim 1, wherein the to-be-passivated side surface of the sheet-like material and the to-be-passivated side surface of the next sheet-like material are located on a same plane.
7. The passivation method according to claim 1, wherein the protection film comprises a film layer with elasticity in a thickness direction to have the grid line of the sheet-like material embedded in the protection film.
8. The passivation method according to claim 7, wherein the protection film further comprises a flexible film layer.
9. The passivation method according to claim 1, wherein a material of the protection film comprises one or more combinations of polyimide, mica polybenzimidazole, and glass fiber.
10. The passivation method according to claim 1, wherein the sheet-like material extends along a plane parallel to a first direction and a second direction, the grid line extends along the first direction, and the to-be-passivated side surface is located at an end of the sheet-like material in the first direction.
11. A cell, wherein the cell is obtained based on a sheet-like material prepared by the passivation method according to claim 1.
12. An auxiliary passivation device for implementing a passivation method, configured to passivate to-be-passivated side surface of stacked sheet-like materials through the passivation method, wherein the sheet-like material is configured to prepare a cell, and at least one of a front surface and a back surface of the sheet-like material comprises a grid line; and the auxiliary passivation device for implementing the passivation method comprises: a carrying device, configured to carry the sheet-like material for bringing it into a passivation device for passivation; and at least one layer of protection film, configured to fill a gap between adjacent sheet-like materials.
13. The auxiliary passivation device for implementing the passivation method according to claim 12, wherein the sheet-like material comprises a to-be-passivated side surface, and a number of the protection film is multiple; and the carrying device comprises: a first flat plate, configured to carry the sheet-like material, and a plurality of protection films are disposed between the first flat plate and the sheet-like material, or between adjacent sheet-like materials respectively.
14. The auxiliary passivation device for implementing the passivation method according to claim 13, further comprising: a second flat plate, disposed opposite to the first flat plate to form an accommodation space, wherein the accommodation space is configured to accommodate the stacked sheet-like materials, and the plurality of protection films are further disposed between the sheet-like material and the second flat plate; and a side plate, detachably connected to the first flat plate and the second flat plate.
15. The auxiliary passivation device for implementing the passivation method according to claim 13, wherein the carrying device further comprises: a box body, comprising an accommodation

chamber, wherein the accommodation chamber is configured to accommodate the first flat plate, the second flat plate, and the sheet-like material and the protection film accommodated between the first flat plate and the second flat plate; and a box cover, detachably connected to the box body.

16. The auxiliary passivation device for implementing the passivation method according to claim 15, wherein a gap is provided between the box body and the box cover when the box body is fastened with the box cover, the gap is configured as an inlet for process gas, and when the passivation device passivates the to-be-passivated side surface, the process gas enters the accommodation chamber through the gap.

17. The auxiliary passivation device for implementing the passivation method according to claim 16, wherein the box body comprises a protrusion on a side, configured to engage with the box cover, of the box body, to create the gap when the box body and the box cover is fastened; and a size of the protrusion in a direction perpendicular to the box cover is equal to a size of the gap in the direction perpendicular to the box cover.

18. The auxiliary passivation device for implementing the passivation method according to claim 15, wherein the sheet-like material further comprises at least one non-passivated side surface; and the box body comprises: a plurality of box boards, and the plurality of box boards is configured to cover the at least one non-passivated side surface of the sheet-like material.

19. The auxiliary passivation device for implementing the passivation method according to claim 11, wherein the sheet-like material is further provided with at least one non-passivated side surface; and the protection film comprises an extension portion, and the extension portion covers at least one non-passivated side surface.

20. The auxiliary passivation device for implementing the passivation method according to claim 19, wherein a size, in a thickness direction of the sheet-like material, of the extension portion is greater than or equal to a thickness of one piece of the sheet-like material.